Sixfold qubit savings with small Bolza codes: a replication of Higgott's quantum coding study

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Abstract

We partly replicated the small-code result of Higgott (2024): Bolza protects four logical qubits using 16 rather than 96 physical qubits, but its confirmed block-failure probability at physical error probability $p = .001$ is .13772 rather than the figure-derived .13390; four planar copies give .12925 rather than .12977. We cannot establish identical performance or replay the original simulations because their run bundle and shot counts were not recovered. Fresh fixed-budget sampling confirms a Bolza–planar difference of .00847, with a simultaneous 95% half-width of .00135 across five selected noise levels. Thus the exact sixfold resource saving accompanies a measured error-rate cost. The original 32-point reconstruction remains available as a baseline: its maximum source difference was .01334, inside an exploratory .02 margin that is not an equivalence criterion. This bounded study covers Figure 7.14a and selected small-code parameters, not the entire dissertation.

Claim	Published	Ours	Direction	Verdict
Physical qubit ratio	6	6	Same	Match
Bolza failure, $p = .001$	.13390	.13772	Higher	Partial
Planar failure, $p = .001$	.12977	.12925	Lower	Close
Bolza minus planar	.00414	.00847	Same	Partial
Embedded distances	2; 3	2; 3	Same	Match

Source: Higgott (2024, pp. 174,177–178, Table 7.2 and Figure 7.14a). Error rates are figure-derived, not printed cells. Ours: fresh one-million-shot experiments per basis. The .00847 difference has simultaneous half-width .00135 for five preselected comparisons. Source-comparison bands include our sampling and vector-extraction uncertainty only; unknown original sampling prevents a two-sample verdict. For the planar source comparison, "close" means the absolute gap .00052 is within two of our standard errors plus extraction bounds (.00098); the corresponding Bolza gap .00382 exceeds its .00054 band. Bonferroni adjustment across the ten confirmation source comparisons leaves those two decisions unchanged. These are conditional numerical checks, not equivalence.

1 What replicated?

The sixfold resource reduction replicated exactly, but equality of performance did not. At $p = .001$, fresh confirmation gives Bolza and planar rates .13772 and .12925, against .13390 and .12977 in Figure 7.14a. The published ordering is preserved, with a larger reconstructed difference. Original-source sampling uncertainty remains unknown. The question is whether a small hyperbolic memory can protect four logical qubits with substantially fewer physical qubits than four planar honeycomb memories. The answer remains favorable for resources, with a measurable performance difference in our reconstruction.

The selected result is a bounded study within a much larger dissertation. Higgott's thesis was submitted on October 1, 2023; UCL catalogs the doctorate as 2024. Chapter 7 studies hyperbolic Floquet codes, whose two-qubit parity measurements change the represented logical operators over time. We read that chapter, the decoder definitions on pp. 41–56, and Appendix E. Larger code families, thresholds, the 48-fold saving, real-time decoding speed, and other chapters are outside this replication. The closely related preprint provides the vector figure used in the independent source extraction (Higgott and Breuckmann, 2023).

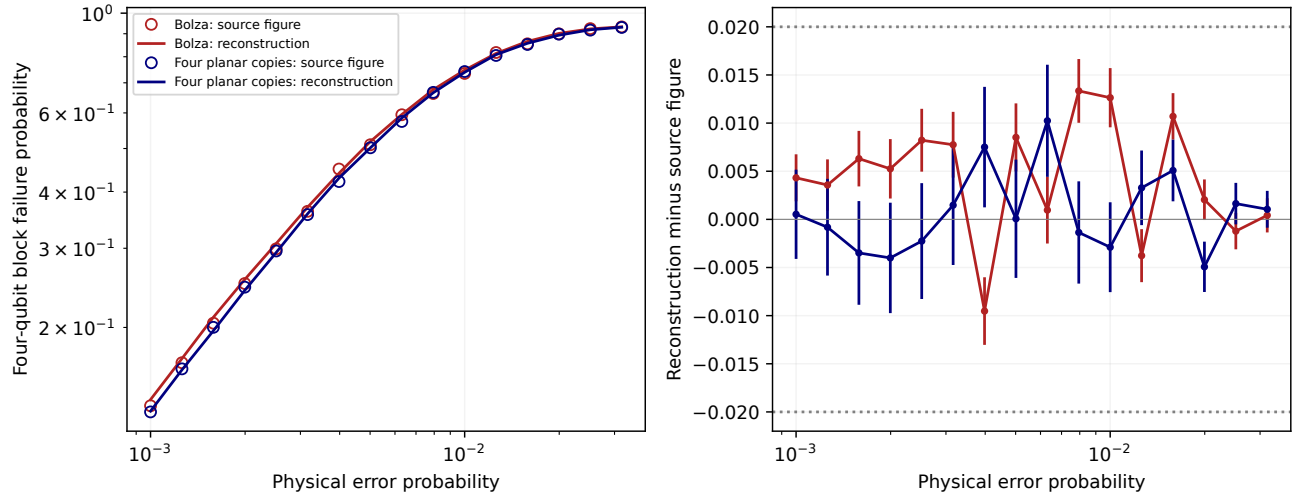


Figure 1: Figure 7.14a reconstructed at 16 physical-noise probabilities. Open circles are independently extracted source-vector coordinates; lines are the preserved baseline of 100,000 shots per basis. Fresh confirmation is reported separately in Table 1. The right panel gives differences with Bonferroni bands for our sampling uncertainty only. Source sampling uncertainty is unknown. The dotted horizontal lines mark the exploratory 0.02 absolute margin. Data: `figure_comparison.csv`.

2 What we did and what came from the source

We independently reconstructed noise, checked code parameters, simulated circuits, and compared the resulting curves. The source supplied the Bolza edge schedules, detector definitions, and observable parities as Stim circuit data; the planar geometry and schedules came from the comparator generator cited by the thesis (Gidney et al., 2022). We did not independently regenerate the hyperbolic tiling from a group presentation. Our parameter check instead reconstructed the three edge matchings and their faces from the supplied noiseless circuits.

Binary linear algebra verifies that both small lattices have Euler characteristic -2 and encode four logical qubits. The 16-qubit lattice has 24 edges and six faces; its 64-qubit refinement has 96 edges and 30 faces. Exhaustive low-weight searches in all three restricted codes recover embedded distances 2 and 3, independently of Stim's circuit-distance routine. For the 64-qubit code, one sector has no logical vector through weight four; the complementary sector has distance three, which suffices to establish the minimum. See `code_parameters.py` and `independent_code_parameters.json`.

Stim's graphlike circuit distances independently agree with those embedded minima for reconstructed EM3 circuits. The supplied 16-qubit SD6 circuits also give distance two and use 40 physical qubits. The $l = 2$ SD6 total of 160 follows from the source's $n+3n/2$ resource formula, but we have not independently reconstructed its SD6 circuit distance four. That cell is not a numerical replication result.

3 Methods and uncertainty

The comparison uses the probability of at least one logical failure, not a mean of four observable error indicators. The thesis defines this convention in Figure 7.11's caption and illustrates independent-patch aggregation in Figure 7.12. If r_X, r_Z are single-patch planar failure probabilities, our four-copy quantity is

$$R_{\text{planar}} = \tfrac{1}{2}\{1-(1-r_X)^4 + 1-(1-r_Z)^4\}.$$

For Bolza, each experiment already encodes four logical qubits; we average its X- and Z-basis block probabilities. The four copies are assumed independent, but the four observables within Bolza are not.

We ran 226 conditions with 100,000 shots each, totaling 22.6 million primary and diagnostic shots. This includes both durations, both planar noise variants, both bases, and all planar comparator sizes for the second Bolza code. Seeds, counts, circuit hashes, and processing times are saved for every condition by `run_replication.py`. Separate historical and generator checks add further samples; these are not silently pooled into the primary estimates.

Our uncertainty calculations separate three sources of error. For an independently sampled basis with empirical rate r and N shots, the estimated variance is $r(1-r)/N$. We propagate the planar transformation with derivative $4(1-r)^3$ and divide the sum of independent basis variances by four when averaging. The two-standard-error bands and Bonferroni bands with multiplier 3.16282 are both saved. The simultaneous construction uses the union bound over 32 comparisons: choosing each two-sided tail probability as .05/32 bounds the probability of any interval miss by .05. This is the single-step Bonferroni calculation, not a step-down procedure (Holm, 1979). These normal approximations are used for the primary high-count curves, not as precise rare-event confidence bounds for the second code's smallest rates.

Vector-coordinate bounds are not statistical error bars. The independent source reviewer allowed $\pm.05$ original PDF points and propagated that interval through the logarithmic axes. All 32 marker centers were independently aligned to the thesis panel, with maximum residual 1.47×10^{-5} thesis PDF points. Original simulation shot counts were not recovered. Consequently, the 16 conditional band exceedances are differences relative to our precision, not 16 demonstrated failures of the original sampling procedure. A two-sample Monte Carlo test is unavailable.

Our practical comparison margin is explicitly exploratory. We fixed an absolute probability margin of .02 after pilots and before the full curve summary, not in a preregistration and not because the source specified it. Every point estimate lies inside this margin relative to the figure. The largest reconstructed Bolza–planar gap is .01638, but its simultaneous uncertainty can extend to .02347. We therefore do not establish uniform equivalence inside .02 over the entire grid.

4 Assumptions supplied in this replication

4.1 A finite grid and finite sampling budget

Our numerical scope is the 16 plotted probabilities from .001 to about .03162 for the first size. The second size is a five-point extension at .001, .002, .005, .01, and .02. Each baseline basis uses a fixed 100,000 shots; zero failures, if present, would not imply zero risk. Finite sampling budgets define our precision, not the original study's uncertainty; seeds and workload details are supplied in the reproduction appendix.

4.2 Modern and historical software

Our primary implementation uses Stim 1.16.0 and PyMatching 2.4.0 because the thesis specifies stabilizer simulation and minimum-weight perfect matching (Gidney, 2021; Higgott, 2021). A separate Python 3.10 environment uses Stim 1.12.0 and PyMatching 2.1.0, representative of the thesis period, on all 64 primary basis circuits. These are sensitivity versions, not a recovered original environment. Python 3.12.14, NumPy 2.5.4, SciPy 1.18.1, NetworkX 3.7, and Matplotlib 3.11.2 support array calculations, uncertainty, independent graph optimization, and figures (Harris et al., 2020; Virtanen et al., 2020; Hagberg et al., 2008; Hunter, 2007; Software projects, 2026). Exact dependencies are in `requirements-lock.txt`.

5 Ambiguities resolved by checks

5.1 Measurement duration

Forty-eight subrounds are the source-consistent duration for Figure 7.14a. Page 157 defines a round as three subrounds, and page 162 describes six subrounds as two rounds. The figure caption says 16 rounds, while the supplied circuit filenames and actual schedules contain 96 subrounds. We reduced the bulk repeat count from 15 to seven, preserving initial and final boundary operations. Both versions were run; the longer circuit substantially raises the low-noise failure probability. The mismatch is an input-selection issue, not evidence that the plotted simulation used the supplied template unchanged. See `circuits.py`: `with_duration`.

5.2 Correlated EM3 events and a simulation helper

The EM3 channel requires a correlated event, not independent depolarization and measurement noise. Section 7.2.5.1 samples uniformly from 32 two-qubit Pauli/measurement-flip possibilities with probability p, including the identity outcome. A channel with all 31 nonzero binary generators applied independently at probability

$$q = \frac{1 - (1 - p)^{1/16}}{2}$$

has the same distribution: every nontrivial character encounters 16 generators, giving $(1 - 2q)^{16} = 1 - p$. The implementation is `circuits.py`: `em3`. Direct convolution agrees within 7.8×10^{-16}. Every one of the 768 event blocks in each supplied EM3 basis circuit contains precisely those 31 generators. Stored $q = .0000312647$ implies $p = .00100000134578$; this rounding is immaterial at our precision. The extra simulation qubit is a noiseless parity-measurement helper, not a seventeenth physical EM3 qubit.

5.3 Boundary noise in the planar comparator

The generator's plotting scripts map the displayed label EM3 to its `EM3_v3` model. An example collection command uses `EM3_v2`; the two models differ at single-qubit boundary measurements. Running both changes the four-copy result at $p = .001$ from .15755 (v2) to .13030 (v3), the latter close to the figure's .12977. We retained all v2 results as a failed reconstruction diagnostic rather than discarding them. This correction follows inspected source code and improves source alignment; it does not constitute fitted calibration of p.

5.4 Logical representatives: what the algebra establishes

The supplied observables can be paired into a canonical basis, but the exact published-path mapping remains unresolved. At two matching bulk phases, ticks 7 and 43, we explicitly invert the four-by-four cross-commutation matrix and construct transformed Z representatives that pair with the supplied X representatives. The transformation and Pauli strings are in `basis_mapping.json`; `basis_mapping.py` regenerates them. For an invertible within-sector map A, $e = 0$ if and only if $Ae = 0$, so consistent relabeling of observed and predicted errors preserves block failure.

This algebra does not identify the illustrated paths in Figure 7.6 with circuit vertex labels, nor independently reconstruct their initial and final readout paths. The source gives the path construction and six-subround evolution (pp. 161–163), while the data README warns that its representatives differ. No explicit figure-to-circuit vertex correspondence was recovered. We therefore retain this as a source-equivalence limitation. Canonical pairing alone is not a proof that a rebuilt decoder makes the same correction decisions.

6 Independent correctness checks

Independent checks support the implementation, while bounding what they verify. A mutually exclusive categorical representation of the noise was compared with the independent-generator representation on ten code/basis arms with 100,000 shots per generator. The largest standardized difference was 2.962, below a Bonferroni-adjusted threshold for ten two-sided comparisons. All ten noiseless circuits produced zero detector and observable events in 4096 samples each. Exact no-correction probabilities computed from the detector-model characteristic function provide another control, independent of decoding (`run_replication.py`: `no_correction_exact`).

A separate NetworkX shortest-path and matching calculation (`independent_checks.py`: `independent_minimum`) checked 100 syndromes per arm. Its minimum objective differed from PyMatching by at most 1.26×10^{-6} across 1000 cases. This checks optimization on the same graph; it does not independently derive that graph or rule out alternative equal-weight logical corrections. The categorical sampler still uses Stim's stabilizer engine. These are complementary checks, not entirely independent end-to-end implementations. Results and algorithms: `independent_checks.py` and `independent_checks.json`.

The external generator's relevant tests pass, but its entire selected suite does not. After one SciPy import-location compatibility change, 270 tests passed and 20 failed; the failures concern SD6/SI1000 decomposed-distance expectations under the modern Stim version. The EM3 subset passes all 122 selected tests. Those unrelated failures remain recorded and were not edited away. The workflow's own 25 tests also pass.

7 What did fresh confirmation and decoder checks change?

Fresh samples sharpen the resource–performance tradeoff without closing every source gap. We fixed five original grid indices and one million new shots per basis before the follow-up, with new seeds and no significance-based stopping. These 20 million shots remain separate from the original baseline. Table 1 reports the resulting probabilities and simultaneous intervals for the five Bolza–planar differences. Every difference is positive under the reconstructed model; no application-specific acceptable loss was supplied, so we report these costs directly rather than assert equivalence under a new margin.

The largest source difference among these selected confirmation points is .01501, for Bolza at $p = .01$ (.74941 versus .73440). Additional samples therefore did not remove that residual. The original

p	Bolza	Four planar copies	Difference	Simultaneous half-width
0.00100	0.13772	0.12925	0.00847	0.00135
0.00398	0.44124	0.42630	0.01494	0.00185
0.00631	0.59584	0.58456	0.01128	0.00175
0.01000	0.74941	0.74040	0.00900	0.00143
0.03162	0.93201	0.92990	0.00211	0.00068

Table 1: Fresh confirmation, one million shots per basis. Difference means Bolza minus four planar copies; half-widths use a normal approximation and Bonferroni correction over five differences. They describe our reconstructed models, not source uncertainty. Data: `followup_summary.json`.

exploratory .02 band remains a description of the baseline analysis; it is not a scientifically established equivalence tolerance. The fixed protocol and selection rationale are in `protocol.md`; `confirm_and_decode.py` contains the sampler and paired checks.

Equal-cost decoder ties exist but have a small observed effect in this diagnostic. Reversing graph-edge insertion changed 508 logical predictions across twenty 10,000-shot batches, all at identical objective value. The largest basis failure-count change was six in 10,000. Adding an invertible observable relabeling produced the same total disagreement count in fifteen eligible batches. This tests graph ordering and label consistency on the same sampled syndromes, not every possible decomposition or historical convention. Full paired results are in `decoder_diagnostics.json`.

An independent Pauli-frame pilot also supports the small-code noise implementation. It propagates binary Pauli errors directly through pair measurements and samples the 32-category EM3 event with NumPy; it does not call Stim's sampler or detector-model builder. All 32 selected X/Z fault injections agreed exactly with Stim's detector and observable predictions. At $p = .001$, its two basis failure rates were .13954 and .13796, compared with fresh Stim rates .14155 and .13663 at 100,000 shots each. Neither difference exceeds two estimated standard errors. Source schedules and detector annotations are still shared, and decoding still uses the same matching graph. This is a differential error-propagation check, not an independent construction of the ideal quantum state. See `independent_frame.py` and `independent_frame_results.json`.

The actual comparator repository pins also permit a stronger historical diagnostic. Using Stim 1.8.0, PyMatching 0.6.0, NumPy 1.22.1, SciPy 1.7.3 and Matplotlib 3.5.1, we executed the original graph-import path including its large dummy edges. A fixed 100,000 shots per basis at $p = .001$ gives Bolza .13869 and four planar copies .13323. These repository pins are not a recovered Higgott environment; only the runtime subset needed for this serialized-circuit check was installed, with NetworkX 2.8.8 supplied because the repository did not pin it. The result does not recover both source values and differs substantially from the modern adapter's planar .10321. Thus that adapter was not a faithful historical-pipeline proxy. See `source_pinned_check.py`, `source_pinned_results.json`, and `source-pinned-requirements-lock.txt`.

8 What remains unresolved

The remaining curve differences cannot yet be assigned to one cause, even after fresh confirmation and independent error propagation. The largest original-baseline modern difference from a source point is .01334; the historical-version maximum is .01346, so a broad change in library vintage does not close the gap. At $p = .001$, historical software gives Bolza .13803 and planar .13017, close to the modern values. The duration correction and planar boundary model resolve much larger gaps, but do not recover every

source marker.

We rank the remaining causes by available evidence rather than claiming an author error. First, the original run manifest, circuits, graph-conversion choices, seeds, and shot counts are missing; that single run bundle would allow an exact procedural comparison and a legitimate two-sample uncertainty calculation. Second, small boundary or tie-handling differences may survive despite matching code distances and graph objectives. Third, source Monte Carlo fluctuation is plausible but cannot be quantified without counts. The historical graph-import policy was also tested on all 64 primary basis circuits using identical samples for the native and alternative decoders. At $p = .001$, its planar four-copy rate is .10321, farther from the figure than the native .13050 in that paired diagnostic; Bolza changes only from .13749 to .13828. This does not close the source gap. The adapter omits artificial weight-10^{10} dummy edges rejected by modern PyMatching; the diagnostic therefore isolates the graph policy rather than reproducing the entire old decoder unchanged. See `graph_conversion_results.json`. The observed vector-extraction bounds and stored channel rounding are much too small to explain the larger residuals. Numerical optimizers and alternative noise-generation representations agree sufficiently well to make a large implementation fault less plausible.

The second size is supporting evidence, not a complete numerical replication of Figure 7.14b. We ran the reconstructed 64-qubit code and all three planar sizes at five noise strengths and both durations. At $p = .001$ and 48 subrounds, rates are .00209 for Bolza, .13030 for four distance-two patches, .00489 for four distance-three patches, and .00056 for four distance-four patches. The 216/64 ratio is exactly 3.375, but those new simulations do not establish pointwise recovery of the second published panel. Nor do we claim to have reproduced the 64-qubit SD6 distance.

9 Code and source availability

The supplied circuit data and comparator source are available locally with their licenses and pinned upstream revisions. The Bolza source revision is `457975f0d6c67b3a505aacf44589d36779f5dcc1`; the comparator revision is `cc33baac44c7831bd643db81d0053f8ec6eae9d8`. The latter includes a documented one-line compatibility patch. The five-page companion `Higgott_Chapter_7_Figure_7_14_Source_Evidence_Appendix.pdf` supplies six reviewed source crops; `README.md` links their files and provenance. The source dissertation PDF was read through Library after the direct local download returned HTTP 403. The full copyrighted dissertation is not bundled. The 600-dpi crops, vector-coordinate CSV, hashes and `extract_figure_points.py` are bundled; re-extraction requires the source PDF bytes identified in that evidence directory.

The dated search ledger distinguishes not recovered from nonexistent. Targeted searches covered the identified GitHub data and comparator repositories, OSF, Zenodo, and GitLab. We did not clone every unrelated repository on the author's account; that workflow departure is recorded. The extended public-history search covered 11 hyperbolic and 52 comparator commits plus the v2 arXiv archive; no original production bundle was found. The publisher supplemental endpoint was inaccessible, not proven absent. The dated findings are in `public_archival_audit.md`. See `availability_and_compliance.json` and `figure_provenance.json`.

10 Next steps

The original Figure 7.14 run bundle and explicit published-path correspondence remain the most useful next inputs. The bounded public archival search did not recover them, and no author contact was made. With its per-basis counts, exact circuits, software versions, and decoder settings, the residual differences

can be separated into sampling fluctuation and procedural differences. Independent vector extraction of Figure 7.14b would extend the second-size comparison, and an independently reconstructed SD6 circuit would test its distance-four cell. Neither missing input is necessary to verify the sixfold integer resource ratio, but both matter for stronger replication claims.

References

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Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70. https://www.jstor.org/stable/4615733. We use the conservative single-step Bonferroni bound, not the step-down refinement; bibliographic metadata and abstract were accessible, not the complete original article.

Software projects. Version-specific packages used for implementation: Python 3.12.14 (https://www.python.org/); NumPy 2.5.4 (https://github.com/numpy/numpy); SciPy 1.18.1 (https://github.com/scipy/scipy); NetworkX 3.7 (https://github.com/networkx/networkx); Matplotlib 3.11.2 (https://github.com/matplotlib/matplotlib). These packages support numerical arrays, normal quantiles, independent graph matching, and plotting respectively. Pytest 9.2.0 (https://github.com/pytest-dev/pytest) runs the checks; pdfTeX 1.40.26 (https://www.tug.org/texlive/) renders the paper. The historical environment uses Python 3.10.22, NumPy 1.24.4, SciPy 1.10.1, NetworkX 2.8.8, and Matplotlib 3.7.3, with its dependency lock included.

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A Artifacts and reproduction

The local package contains the runnable inputs, code, outputs, and editable manuscript. The starting master revision `ca1d06209b03cad666db0a2ffb9678abc89a6b71` records workflow provenance only; it does not contain this new study. The precise local snapshot and commands are documented in `README.md`. Repository links are intended master navigation paths and resolve there after merge. Draft PR 57 supplies current-commit review links. The README identifies immutable baseline and reproducible-code revisions separately.

Claim or task	Evidence and regeneration
Resource and embedded distance	`code_parameters.py`; run with `--source inputs/bolza`.
Curve simulations	`run_replication.py`; fixed shots and seed rules in README, "Rerun simulations."
All source comparisons	`figure_comparison.csv`; regenerate with `summarize.py`.
Initial supplied-data pilot	`audit_supplied_circuits.py`; explicit source and output arguments in README.
Independent checks	`independent_checks.py`; `independent_checks.json`.
Historical version sensitivity	`historical_check.py`; `historical_results.json`.
Graph import sensitivity	`graph_conversion_check.py`; `graph_conversion_results.json`.
Source estimates	`figure_7_14a_vector_values.csv`; supplied source extraction; raw shot arrays absent.
Software and compatibility	`requirements-lock.txt`; `honeycomb_scipy_compat.patch`.
Follow-up checks	`protocol.md`; `confirm_and_decode.py`; README, Follow-up checks.
Source evidence	`README.md`; `extract_figure_points.py`; source PDFs required for re-extraction.
Paper rebuild	`sixfold_qubit_savings__a_replication_of_higgott_bolza_codes.tex`; two `pdflatex` passes from the paper directory.

B Numerical comparisons and reconstruction scope

Each row compares a source-vector estimate with our calculation. Numerical "close" uses the explicitly stated .02 absolute margin; five displayed decimals describe our presentation, not five-decimal precision claimed by the dissertation. Every row has limited reconstruction scope because the original sampling stream and configuration were not obtained. The reason for every "close" label is that its absolute difference is below .02. Conditional statistical exceedances remain separately recorded in `figure_comparison.csv`.

Code	p	Source	Ours	Difference	Precision	Numerical	Scope
Planar	0.00100	0.12977	0.13030	+0.00053	5 d.p.	close	limited
Planar	0.00126	0.16177	0.16096	-0.00081	5 d.p.	close	limited

Code	p	Source	Ours	Difference	Precision	Numerical	Scope
Planar	0.00158	0.20011	0.19663	-0.00349	5 d.p.	close	limited
Planar	0.00200	0.24582	0.24182	-0.00400	5 d.p.	close	limited
Planar	0.00251	0.29586	0.29361	-0.00225	5 d.p.	close	limited
Planar	0.00316	0.35648	0.35795	+0.00147	5 d.p.	close	limited
Planar	0.00398	0.42221	0.42971	+0.00751	5 d.p.	close	limited
Planar	0.00501	0.50187	0.50193	+0.00006	5 d.p.	close	limited
Planar	0.00631	0.57430	0.58454	+0.01024	5 d.p.	close	limited
Planar	0.00794	0.66657	0.66521	-0.00136	5 d.p.	close	limited
Planar	0.01000	0.74158	0.73869	-0.00289	5 d.p.	close	limited
Planar	0.01259	0.80523	0.80851	+0.00328	5 d.p.	close	limited
Planar	0.01585	0.85187	0.85695	+0.00508	5 d.p.	close	limited
Planar	0.01995	0.89816	0.89324	-0.00493	5 d.p.	close	limited
Planar	0.02512	0.91675	0.91838	+0.00164	5 d.p.	close	limited
Planar	0.03162	0.92956	0.93061	+0.00105	5 d.p.	close	limited
Bolza	0.00100	0.13390	0.13822	+0.00432	5 d.p.	close	limited
Bolza	0.00126	0.16695	0.17052	+0.00357	5 d.p.	close	limited
Bolza	0.00158	0.20442	0.21073	+0.00631	5 d.p.	close	limited
Bolza	0.00200	0.25045	0.25570	+0.00525	5 d.p.	close	limited
Bolza	0.00251	0.29911	0.30734	+0.00822	5 d.p.	close	limited
Bolza	0.00316	0.36249	0.37024	+0.00775	5 d.p.	close	limited
Bolza	0.00398	0.45012	0.44060	-0.00952	5 d.p.	close	limited
Bolza	0.00501	0.50980	0.51831	+0.00851	5 d.p.	close	limited
Bolza	0.00631	0.59439	0.59535	+0.00096	5 d.p.	close	limited
Bolza	0.00794	0.66262	0.67596	+0.01334	5 d.p.	close	limited
Bolza	0.01000	0.73440	0.74704	+0.01264	5 d.p.	close	limited
Bolza	0.01259	0.81753	0.81376	-0.00377	5 d.p.	close	limited
Bolza	0.01585	0.85466	0.86535	+0.01070	5 d.p.	close	limited
Bolza	0.01995	0.89946	0.90150	+0.00204	5 d.p.	close	limited
Bolza	0.02512	0.92386	0.92264	-0.00122	5 d.p.	close	limited
Bolza	0.03162	0.93202	0.93244	+0.00042	5 d.p.	close	limited

C Workflow compliance and availability

The following rows distinguish implemented checks from remaining limitations. Passing helper tests alone does not establish manuscript or scientific compliance.

Requirement	Status	Evidence or limitation
answer-first manuscript, headline numbers and tolerance	followed	Fresh confirmation and resource/risk tradeoff lead; .02 retained only as historical exploratory baseline margin.
R/lavaan/OpenMx engine	not applicable	quantum stabilizer sampling and graph matching, not SEM; sem_preflight.json retained
SEM fit-index comparability fields	not applicable	comparability instead records noise model, duration, physical resources, basis aggregation, decoder and source availability
all selected headline arms	followed	226 checkpoint files; both bases and all source comparator sizes on documented grids
pilot and resource budgets	followed	environment.json, supplied_circuit_pilot.json; four-core quota,16 GiB; two workers,8192-shot batches
independent parameter/channel/optimizer checks	departed	Verification is partial, not an independently reconstructed end-to-end study. revision2/independent_frame_results.json: independent error propagation and 32 fault checks; shared schedules, annotations and decoder graph remain. Exact published-path mapping unresolved in basis_mapping.json.
historical software and alternate graph policy	followed	historical_results.json and graph_conversion_results.json; exact original environment remains unknown
source crops	followed	source-evidence/README.md, six crops, appendix, extraction program and hashes are bundled.
clone every author repository	departed	original circuit data were found in the directly identified repository; unrelated repositories were not exhaustively cloned; no global unavailability claim is made targeted dated availability entries
scholarly citations and source support	followed	citation_support.json corrects retrieved-passage limits; bibliography has software scholarship and installed version records.
master links and descriptive artifact names	followed	Master navigation links in paper; current-commit PDF/source/evidence review links in draft PR57.
original full simulation stream recovery	departed	original shot counts, manifest and exact run environment not recovered; reconstruction scope explicitly limited revision2/public_archival_audit.md: no production bundle recovered; not evidence none exists.
whole-dissertation replication	not applicable	selected Figure7.14a study only; second-size outputs explicitly supporting; no full-thesis claim
preserve unsuccessful diagnostics	followed	20 broad-suite failures, v2 and96-subround outputs, historical and graph-import alternatives retained
publication	followed	Draft research PR57 authorized; primary PDF and source-evidence directory included. No merge authorized.
Approved layout and rendered manuscript review	followed	Kupzyk preamble at ca1d062 reused; scalable fonts, numeric headline columns and manuscript-specific review.
Copy ledgers into manuscript	followed	build_ledgers.py generates requirement/status/evidence and dated availability tables.
Preserve primary baseline and qualify confirmation	followed	Original results unchanged at 8cbde40; revision2 fixed-budget follow-up reported separately.

Date (UTC)	Source	Result
2026-10-30	UCL institutional record and thesis	Selected chapter and relevant definitions read through Library; direct local request returned403; full dissertation not bundled. Crops and extraction program now in source-evidence.
2026-10-30	hyperbolic-floquet-data GitHub	cloned pinned commit; noisy l=1 and noiseless l=1/l=2 circuits plus parameters recovered; selected inputs bundled
2026-10-30	honeycomb-boundaries GitHub	cloned pinned commit; generator and license bundled with one-line SciPy import patch; executed after user authorization
2026-10-30	arXiv 2308.03750v1 archive	Source reviewer recovered matching vector figures; no original counts. Bundled extraction program requires external PDF source bytes.
2026-10-30	OSF, Zenodo, GitLab	targeted web searches for Higgott Floquet material did not recover thesis-specific shot arrays; this does not establish their nonexistence
2026-10-30	author GitHub account exhaustive clone	not performed; focused on identified relevant source repositories
2026-10-30	Hyperbolic reachable history, tags and releases	11 commits,516 paths; no figure run bundle. No tags/releases; public_archival_audit.md.
2026-10-30	Planar reachable history and release	52 commits across two branches,58 paths; no shot arrays. v1-arXiv release concerns planar paper, not Higgott; repository pins recovered.
2026-10-30	arXiv v2 source and related owner trees	25 archive members; no numeric arrays. Five related repository HEAD indexes screened; not exhaustive histories.
2026-10-30	APS supplemental endpoint	Web retrieval inaccessible; absence not established. No identified alternative bundle from targeted searches.